ECE 340: Semiconductor Devices

Lecture 4: Crystal Structure & Bonding forces and Energy Bands in Solids

Prof. Hyunseok Kim

Lecture (Section C): M,W,F / 2:00-2:50pm / 3017 ECEB

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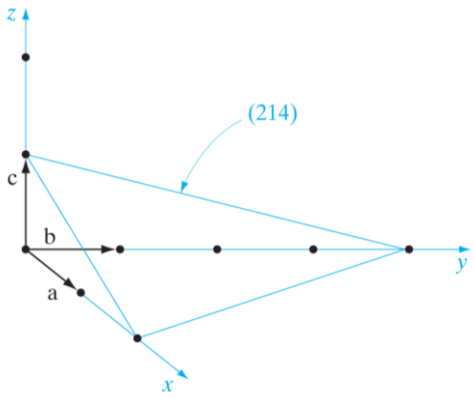
Office Hours: Thursdays 12p-1p, 2256 HMNTL

Recap

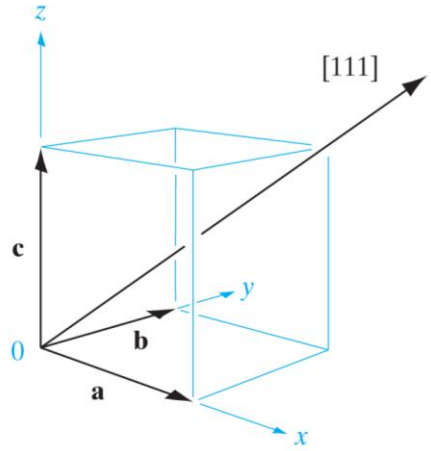
- **Miller Index**

Crystallographic Notation
(hkl): Plane
[hkl]: Vector Normal to Plane
{hkl}: Family of Planes
<hkl>: Family of Vectors

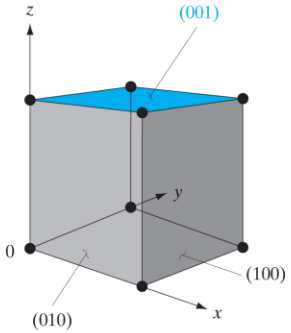
Crystal plane

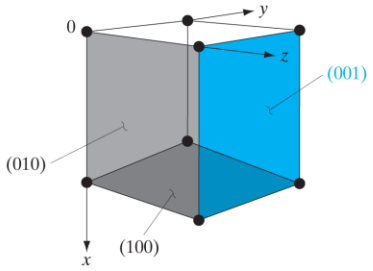


Crystal direction (vector)



- **"Family" of planes/vectors**
  - If related by symmetry operation
  - E.g.) For Si, (100), (010), (001) planes belong to {100} family



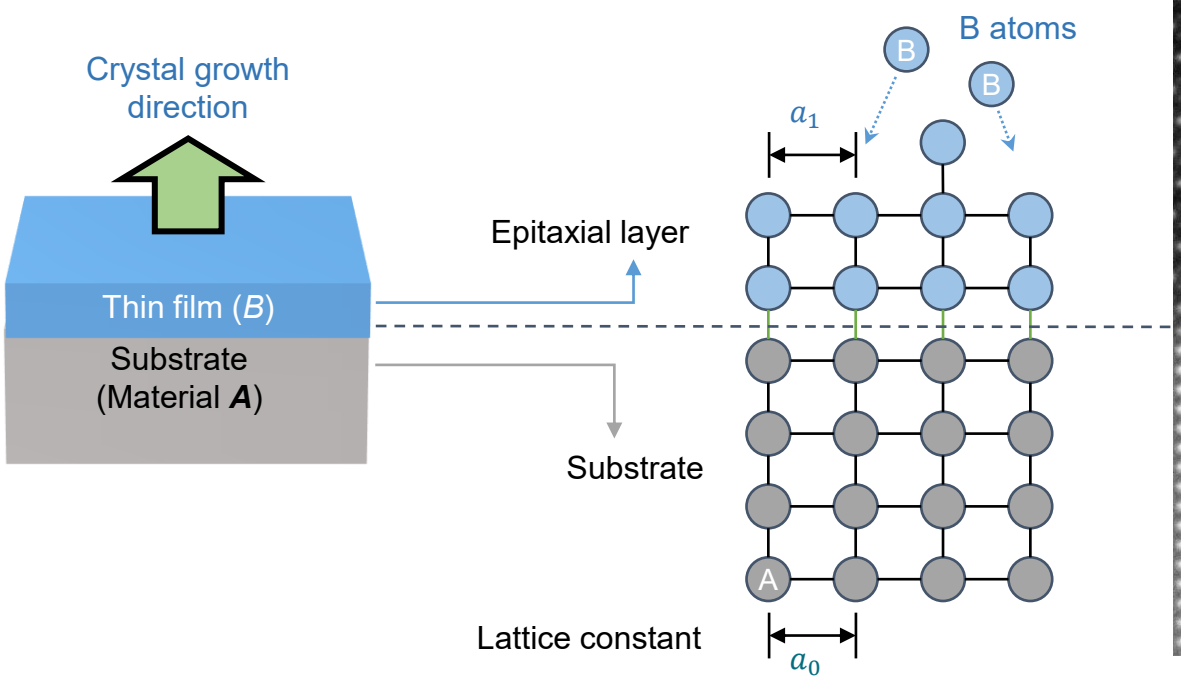


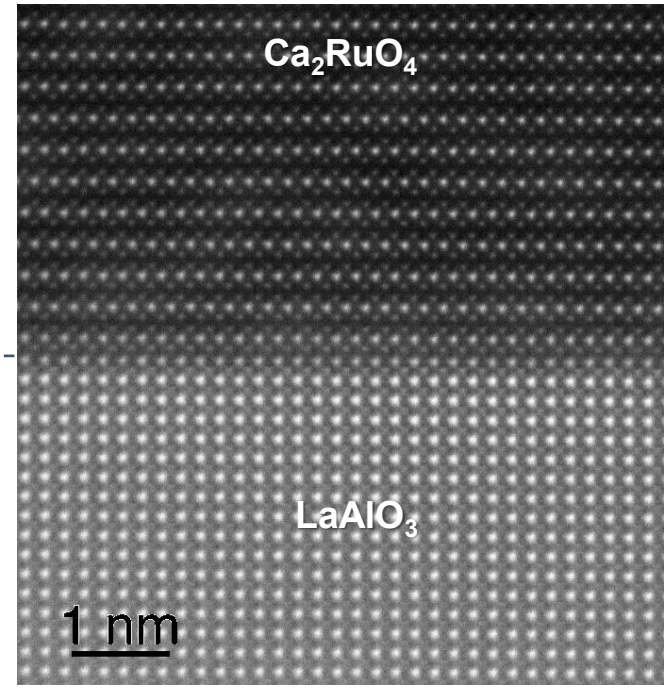
Epitaxial Growth

Epitaxial Growth

Epitaxy: “Crystal growth” on single-crystalline substrates

→ Lattice of grown materials follows the substrate



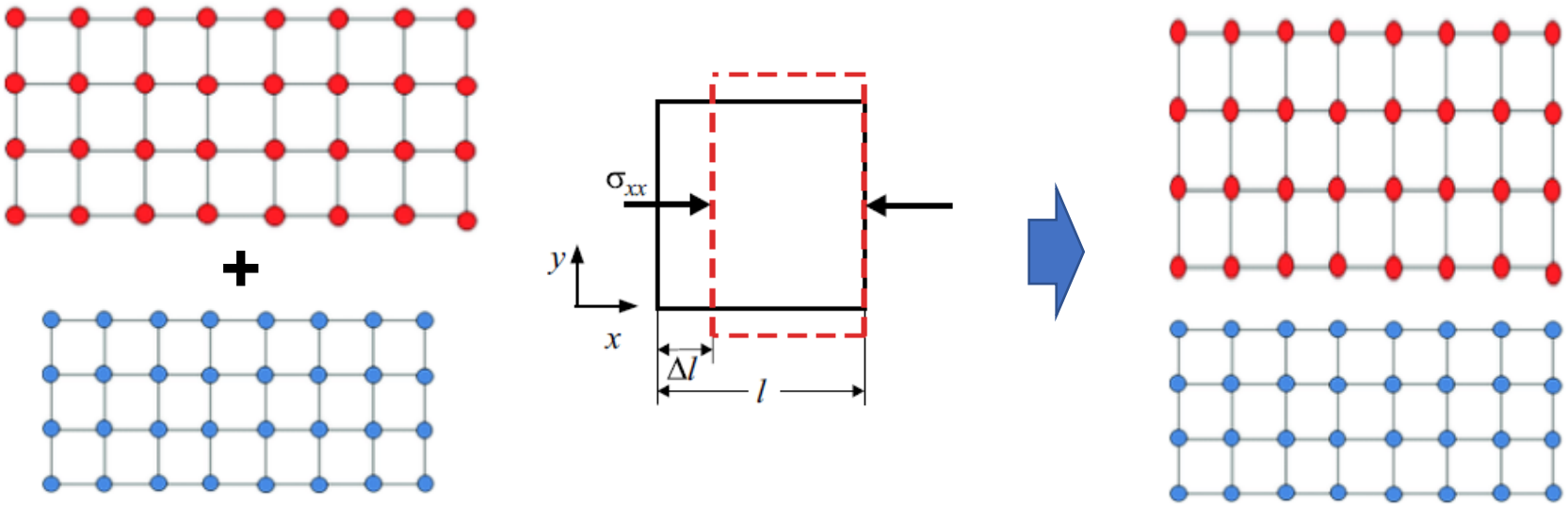


Strain in Epitaxial Growth

What if $a_0 \neq a_1$?

Layer lattice try to "accommodate" to substrate lattice

→ Material is strained



e.g.) Silicon epitaxy on silicon is 'unstrained'
Ge or SiGe epitaxy on silicon is 'strained'

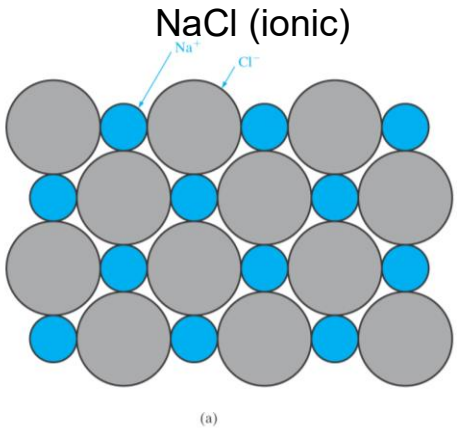
Epitaxial Growth Methods

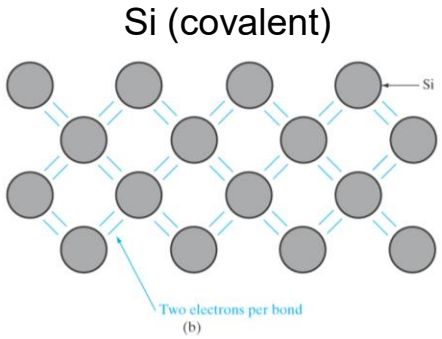
- Liquid Phase Epitaxy
- Vapor Phase Epitaxy
  - (Tri)Chloride & Hydride VPE
- Molecular Beam Epitaxy
- Chemical Vapor Deposition
- Metalorganic Chemical Vapor Deposition

Bonding Forces and Bond Types in Solids

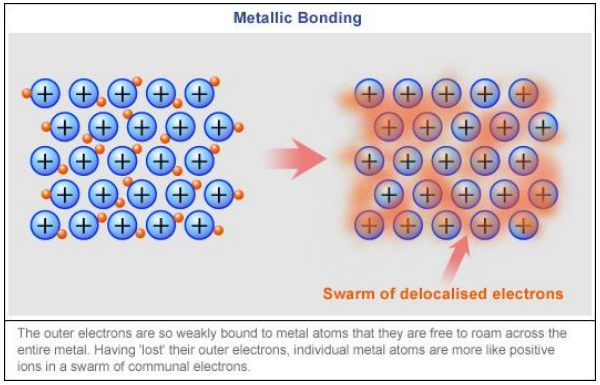
Bond Types

- Ionic Bonding
  - Electron transfer
  - Coulomb attraction balanced by repulsion of nuclei
- Covalent Bonding
  - Electron sharing
  - Si, Ge, C
- Mixed Ionic-Covalent Bonding
  - Electronegativity
  - Polar covalent bonds
  - GaAs, InP, GaN
- Metallic Bonding
  - Positive cores in an electron sea
  - Typical for atoms with 3 or less valence electrons









http://www.abc.net.au/science/articles/2010/04/14/2872429.htm

Population of Atomic Orbitals

Atomic number (Z)	Element	n = 1, l = 0: 1s	2: 0 1: 2s 2p	3: 0 1 2: 3s 3p 3d	4: 0 1: 4s 4p	Shorthand notation
		Number of electrons				
1	H	1				$1s^1$
2	He	2				$1s^2$
3	Li	helium core, 2 electrons	1			$1s^2\ 2s^1$
4	Be		2			$1s^2\ 2s^2$
5	B		2 1			$1s^2\ 2s^2\ 2p^1$
6	C		2 2			$1s^2\ 2s^2\ 2p^2$
7	N		2 3			$1s^2\ 2s^2\ 2p^3$
8	O		2 4			$1s^2\ 2s^2\ 2p^4$
9	F		2 5			$1s^2\ 2s^2\ 2p^5$
10	Ne		2 6			$1s^2\ 2s^2\ 2p^6$
11	Na	neon core, 10 electrons		1		[Ne] $3s^1$
12	Mg			2		[Ne] $3s^2$
13	Al			2 1		[Ne] $3s^2\ 3p^1$
14	Si			2 2		[Ne] $3s^2\ 3p^2$
15	P			2 3		[Ne] $3s^2\ 3p^3$
16	S			2 4		[Ne] $3s^2\ 3p^4$
17	Cl			2 5		[Ne] $3s^2\ 3p^5$
18	Ar			2 6		[Ne] $3s^2\ 3p^6$

Silicon Core and Valence Electrons

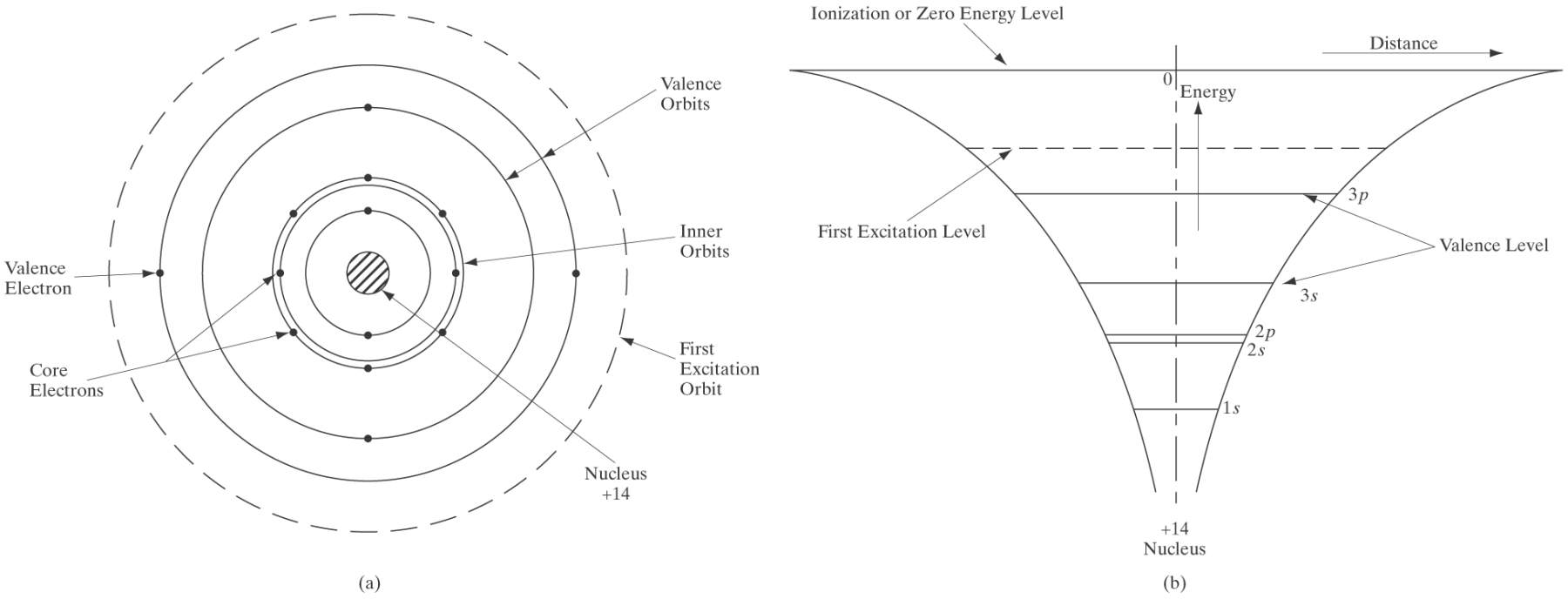
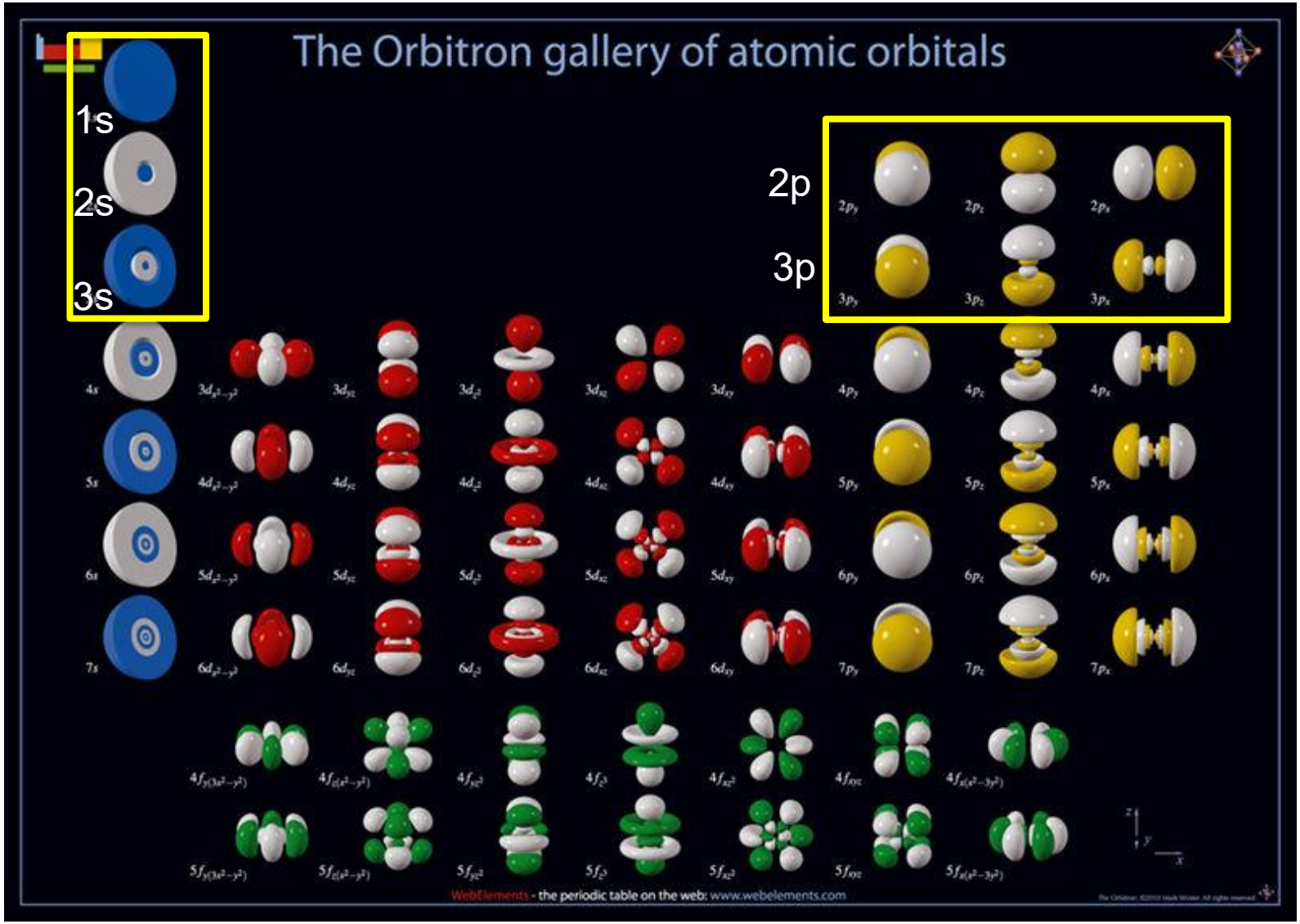


Figure 2.8

Electronic structure and energy levels in a Si atom: (a) The orbital model of a Si atom showing the 10 core electrons (**n** = 1 and 2), and the 4 valence electrons (**n** = 3); (b) energy levels in the coulombic potential of the nucleus are also shown schematically.

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Orbital Wave Function Shape



http://winter.group.shef.ac.uk/orbitron/

sp^3 Hybridization is Si "crystal lattice"

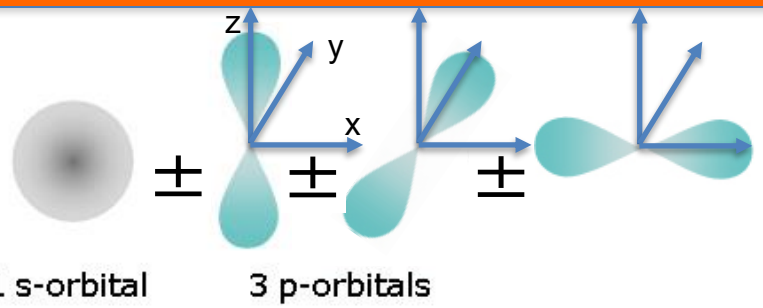
- The sp^3 hybrids are obtained by adding and subtracting all combination of s and p wavefunctions

$$\psi_1 = \frac{1}{2}\left(\psi_s + \psi_{px} + \psi_{py} + \psi_{pz}\right)$$

$$\psi_2 = \frac{1}{2}\left(\psi_s - \psi_{px} - \psi_{py} + \psi_{pz}\right)$$

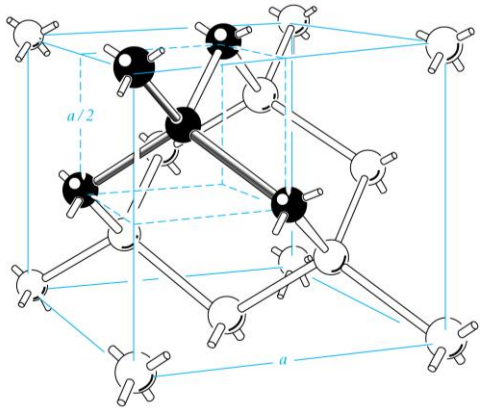
$$\psi_3 = \frac{1}{2}\left(\psi_s + \psi_{px} - \psi_{py} - \psi_{pz}\right)$$

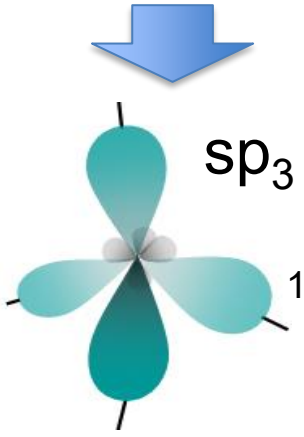
$$\psi_4 = \frac{1}{2}\left(\psi_s - \psi_{px} + \psi_{py} - \psi_{pz}\right)$$





Recall
Si crystal structure





sp_3

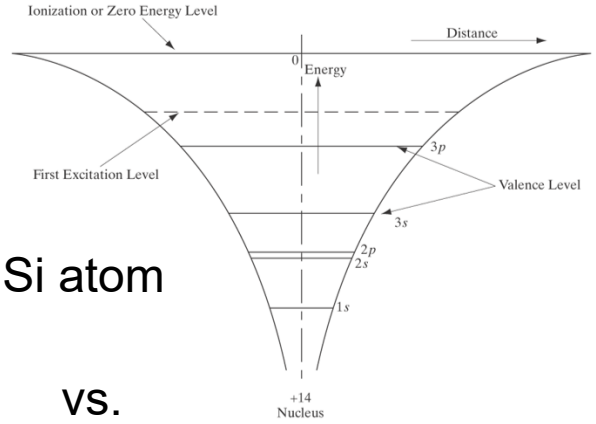
109.5° Bond Angle

http://www.ntu.ac.uk/cels/molecular_geometry/hybridization/Sp3_hybridization/index.html

Energy Bands

General Comments

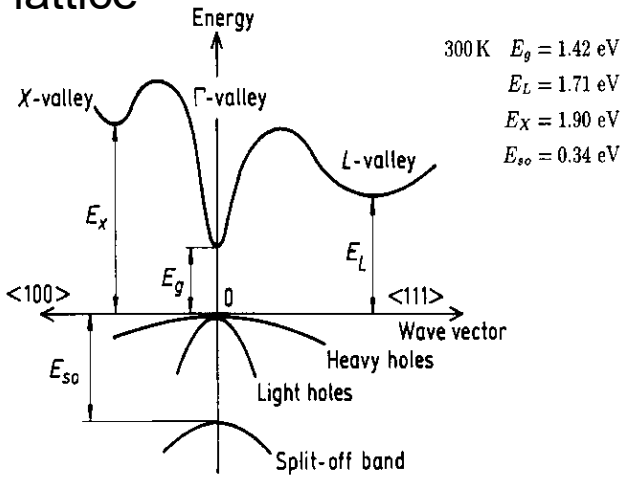
- Streetman's book, and this course, provide a qualitative understanding of how energy bands form
- A detailed understanding of how energy bands form is beyond the scope of this course
  - Energy bands arise as a solution to Schrödinger's Equation when one considers the periodic potential of the lattice



Si atom

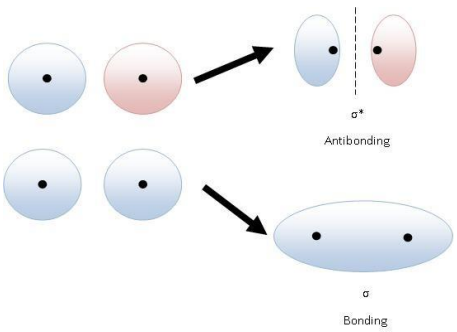
vs.

Si lattice



Bonding and Antibonding Orbitals in Hydrogen (The Simplest Bandgap)

Elemental Hydrogen: $1s^1$
Hydrogen #1: 2 states, 1 electron
Hydrogen #2: 2 states, 1 electron
H_2: 4 states, 2 electrons



http://chemwiki.ucdavis.edu/Theoretical_Chemistry/Chemical_Bonding/Molecular_Orbital_Theory

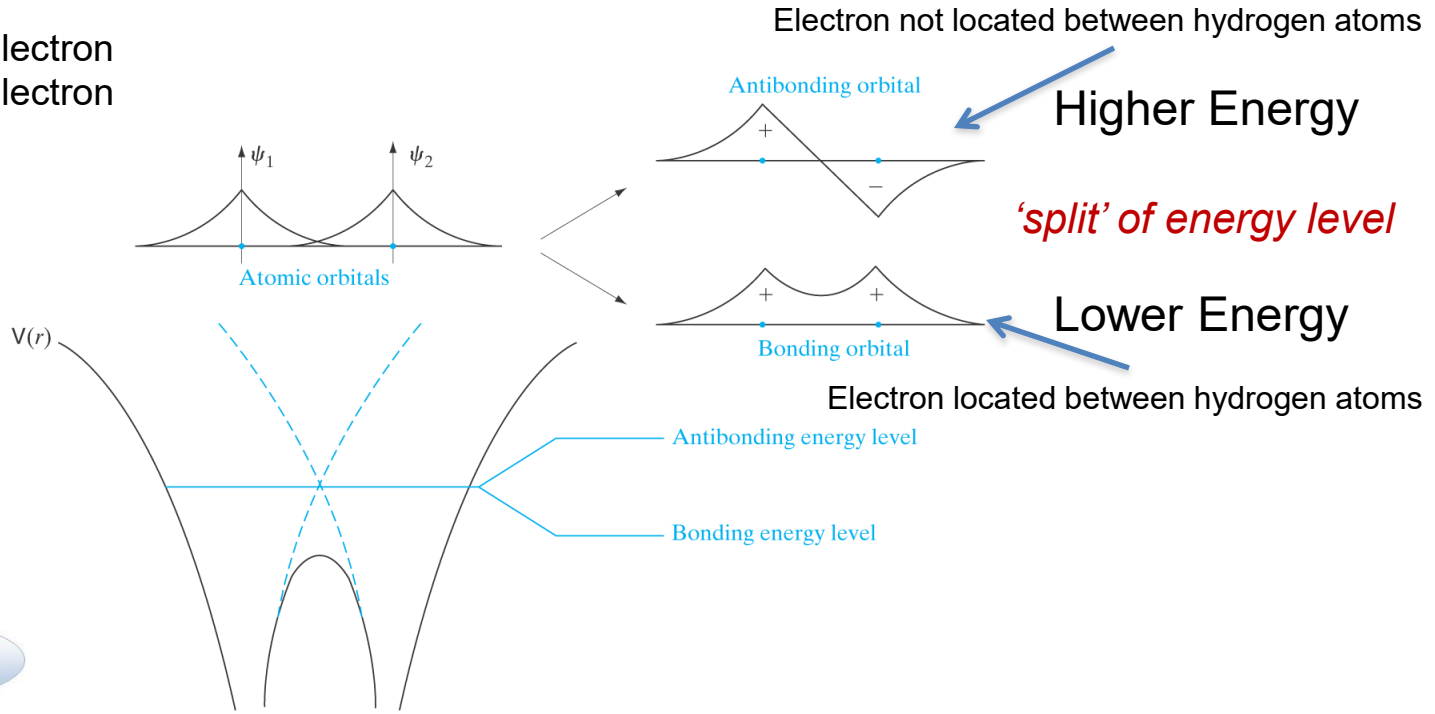


Figure 3.2

Linear combinations of atomic orbitals (LCAO): The LCAO when 2 atoms are brought together leads to 2 distinct "normal" modes—a higher energy antibonding orbital, and a lower energy bonding orbital. Note that the electron probability density is high in the region between the ion cores (covalent "bond"), leading to lowering of the bonding energy level and the cohesion of the crystal. If instead of 2 atoms, one brings together *N* atoms, there will be *N* distinct LCAO, and *N* closely spaced energy levels in a band.

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Energy Bands

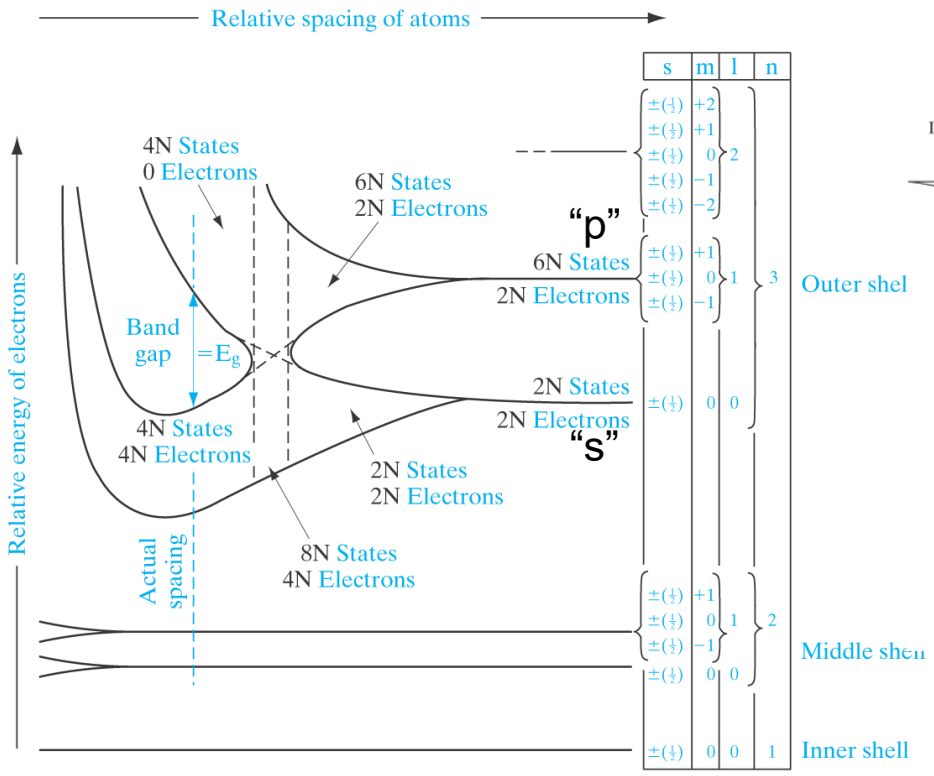
- As atoms are brought together to form a solid, the forces of attraction and repulsion between atoms will find a balance at the proper inter-atomic spacing for the crystal
- **As the atoms are brought closer together, the discrete atomic energy levels form "bands".** These bands become the conduction and valence band, separated by the forbidden gap where no electron states exist.
- Silicon Atom: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^2$ (missing 4 electrons to fill outer shell)
  - Core: $1s^2\ 2s^2\ 2p^6$ Valence: $3s^2\ 3p^2$

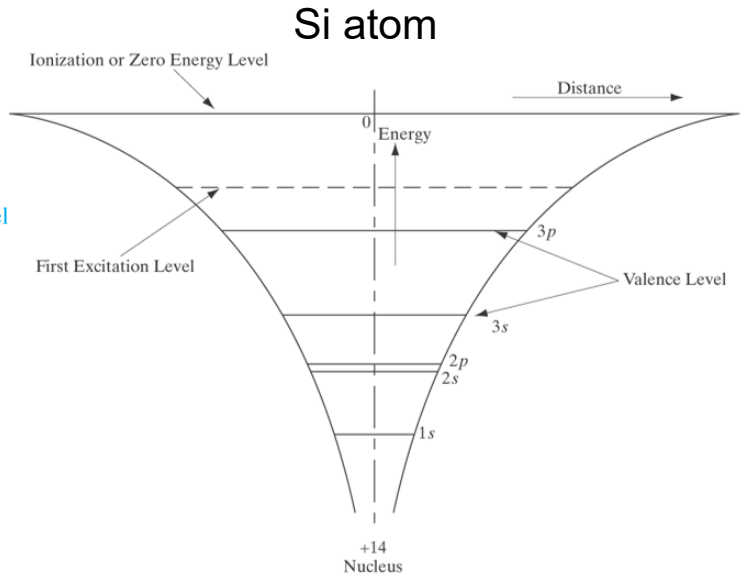
n=3	Available States	# of Electron
3S	2	2
3P	6	2
Total	8	4

- Assuming we have N Si atoms to form a crystal, we have N x (3s2 3p2) valence electrons

n=3	Available States	# of Electron
Total	Nx8	Nx4
Conduction Band	Nx4	0 (at 0° K)
Valence Band	Nx4	Nx4 (at 0° K)

What happens when a collection of Si atoms are brought closer together?





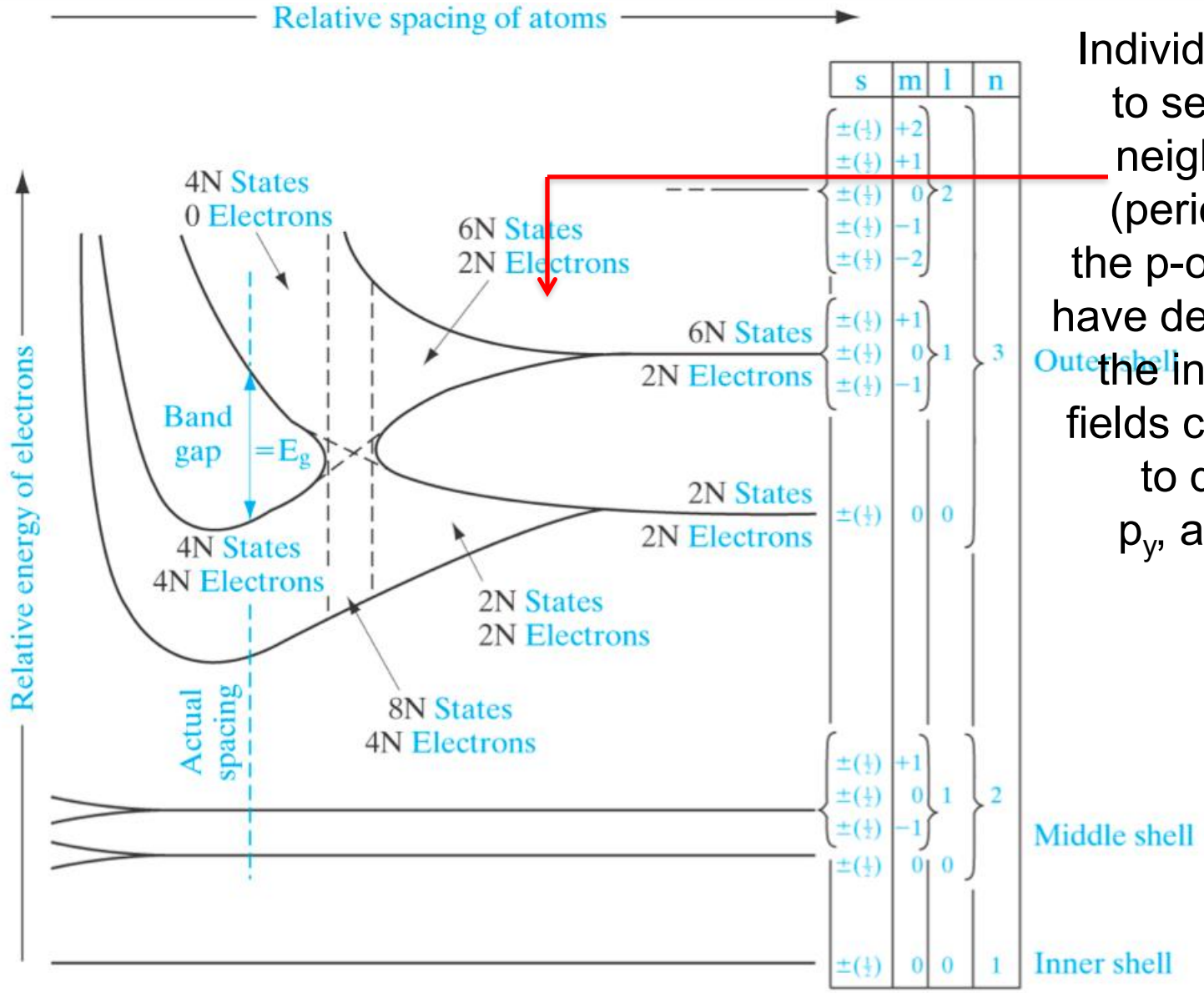
Smaller Lattice Spacing ← Larger Spacing: Isolated Atoms

Figure 3.3

Energy levels in Si as a function of interatomic spacing. The core levels (**n** = 1,2) in Si are completely filled with electrons. At the actual atomic spacing of the crystal, the 2*N* electrons in the 3*s* subshell and the 2*N* electrons in the 3*p* subshell undergo sp^3 hybridization, and all end up in the lower 4*N* states (valence band), while the higher-lying 4*N* states (conduction band) are empty, separated by a band gap.

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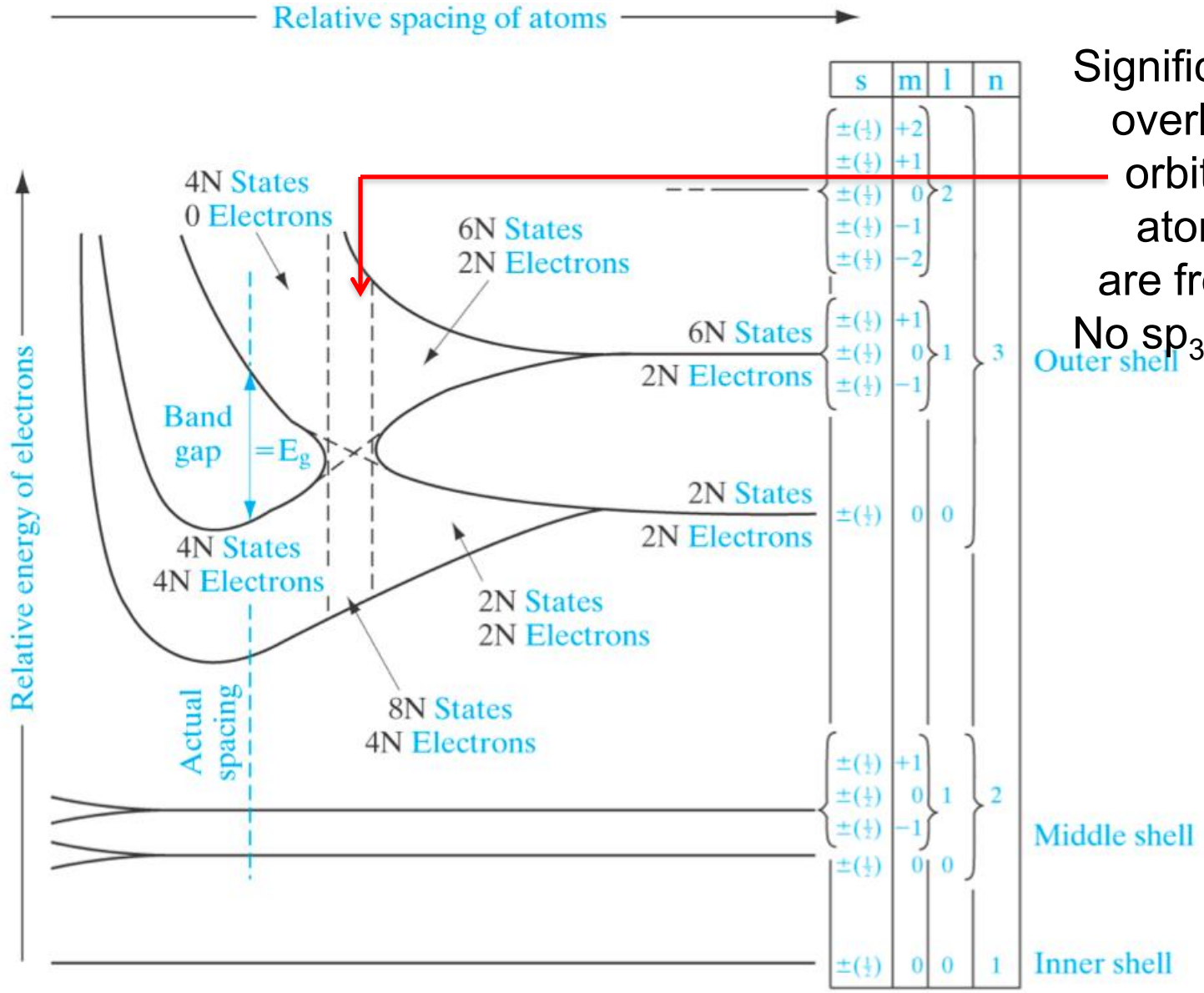
Large Spacing: Weak Potential



Individual atoms begin to see the influence neighboring atoms (periodic potential), the p-orbitals no longer have degenerate energy: the interaction of the fields cause differences to develop in p_x, p_y, and p_z orbitals*

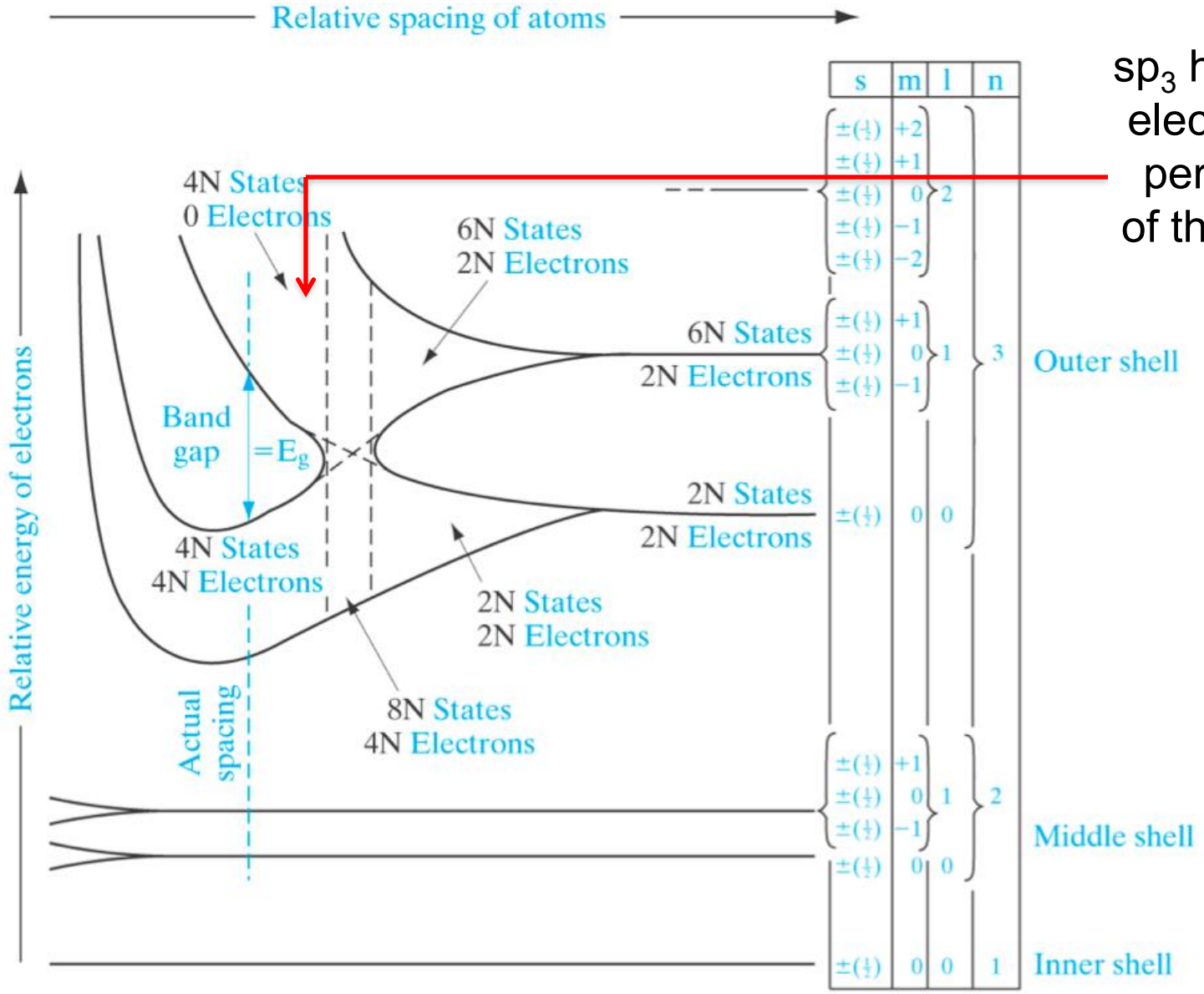
*Simplified picture

Wave Function Overlap Region



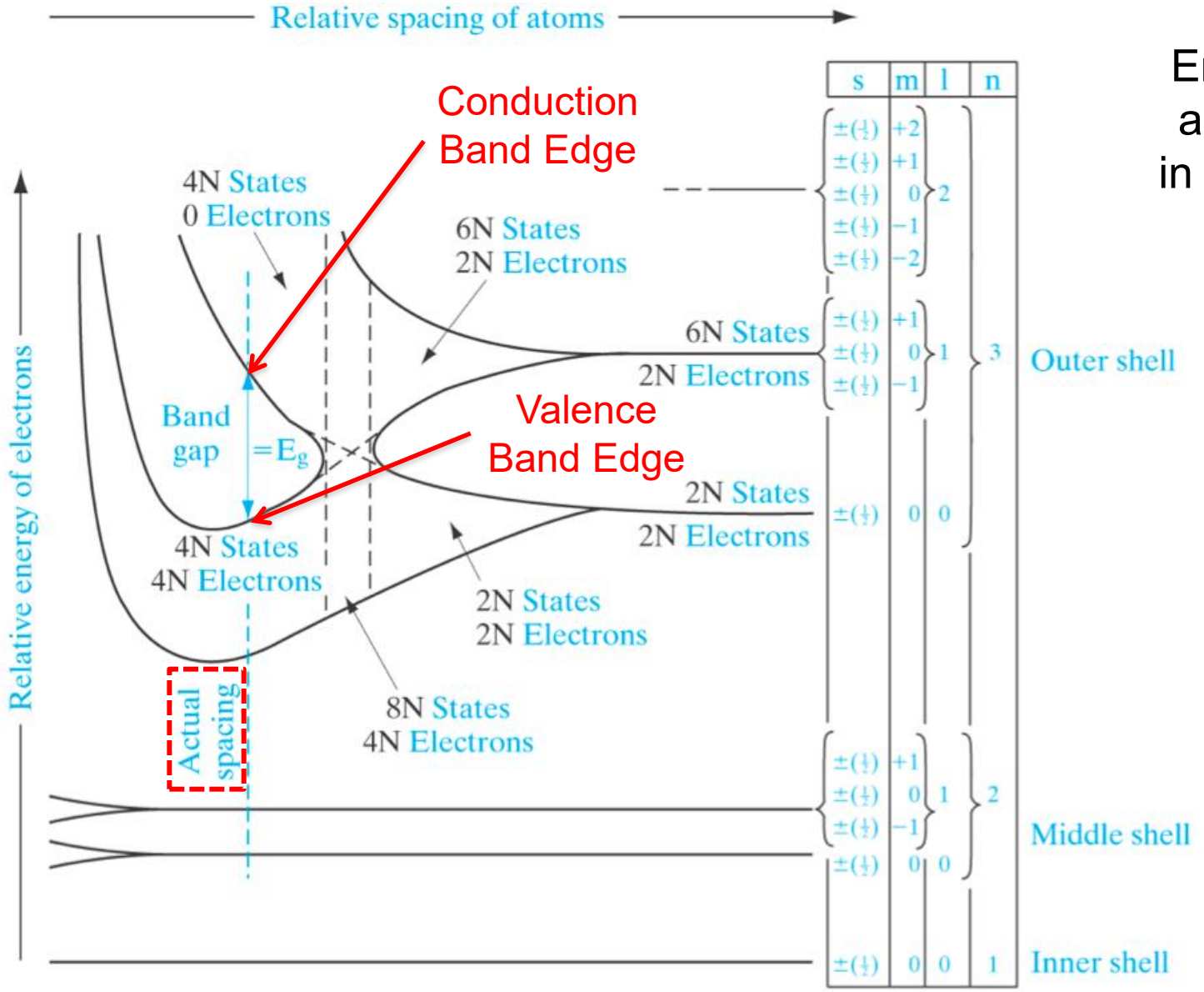
Significant wavefunction overlap between the orbitals of adjacent atoms – electrons are freely exchanged. No sp_3 hybridization yet.

sp_3 Hybridization and Lattice Periodic Potential



sp_3 hybrid form, and electrons "feel" the periodic potential of the crystal lattice

Formation of Energy Bands



Energy bands and bandgap in silicon lattice

Depiction of Bands in Real Space

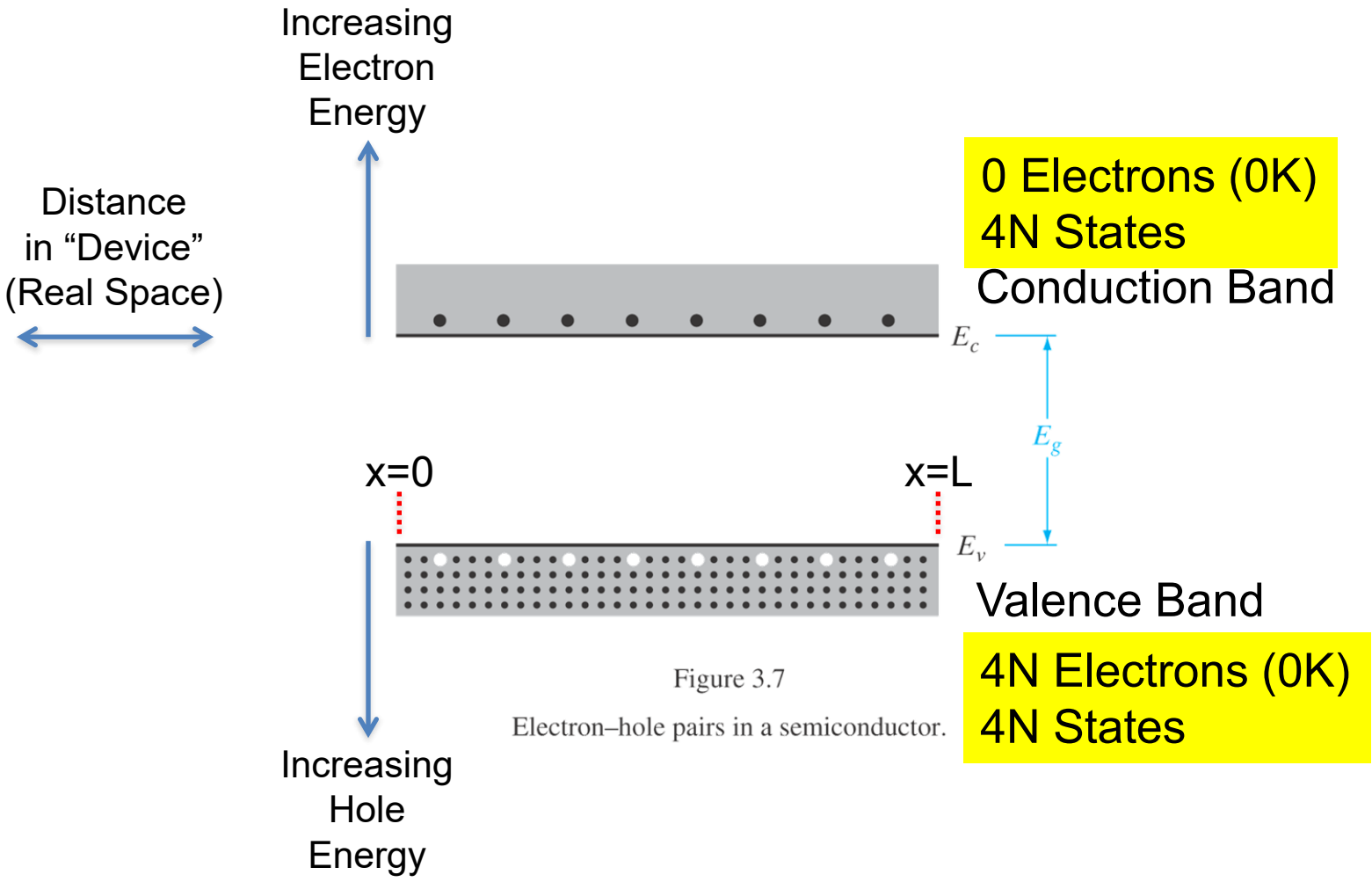


Figure 3.7

Electron–hole pairs in a semiconductor.

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Electron energy-momentum in free space

- Particle's momentum ***p***

$$p=mv \rightarrow v=p/m$$

- Kinetic energy ***E*** of a particle

$$E = \frac{mv^2}{2} = \frac{1}{2}\left(\frac{p}{m}\right)^2 = \frac{p^2}{2m}$$

- De Broglie relation

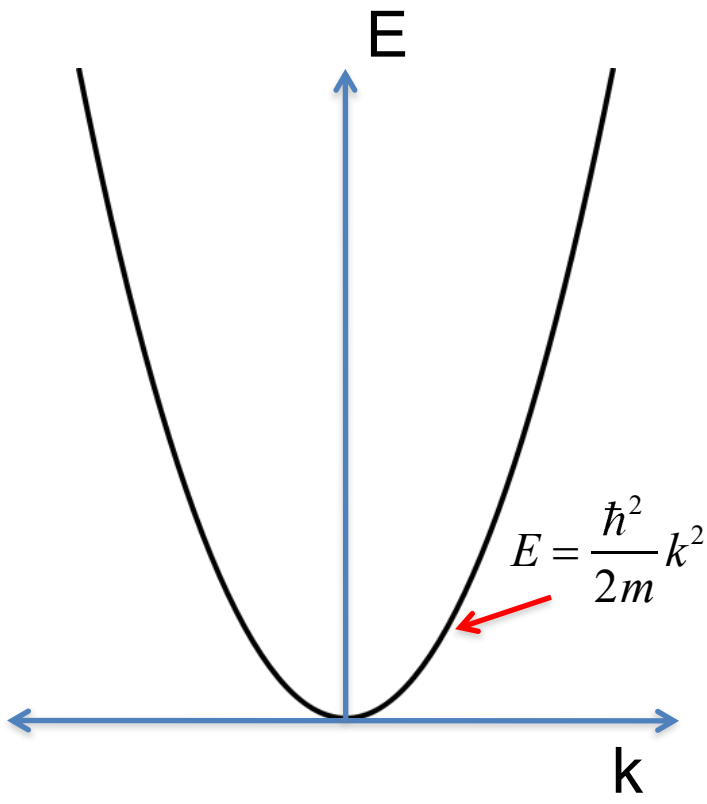
$$\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p}$$

- Define wavenumber ***k***

$$k = \frac{2\pi}{\lambda} \quad \text{, then } p = \hbar k$$

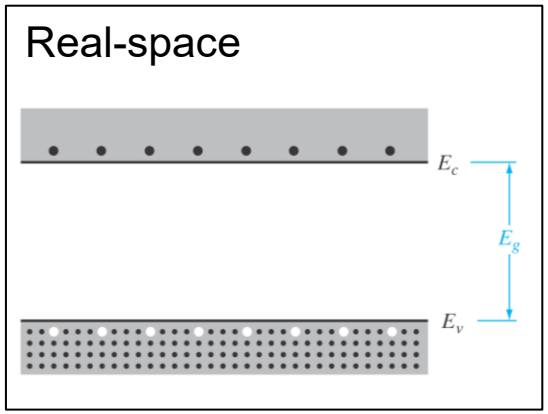
$$E = \frac{p^2}{2m} = \frac{\hbar k^2}{2m}$$

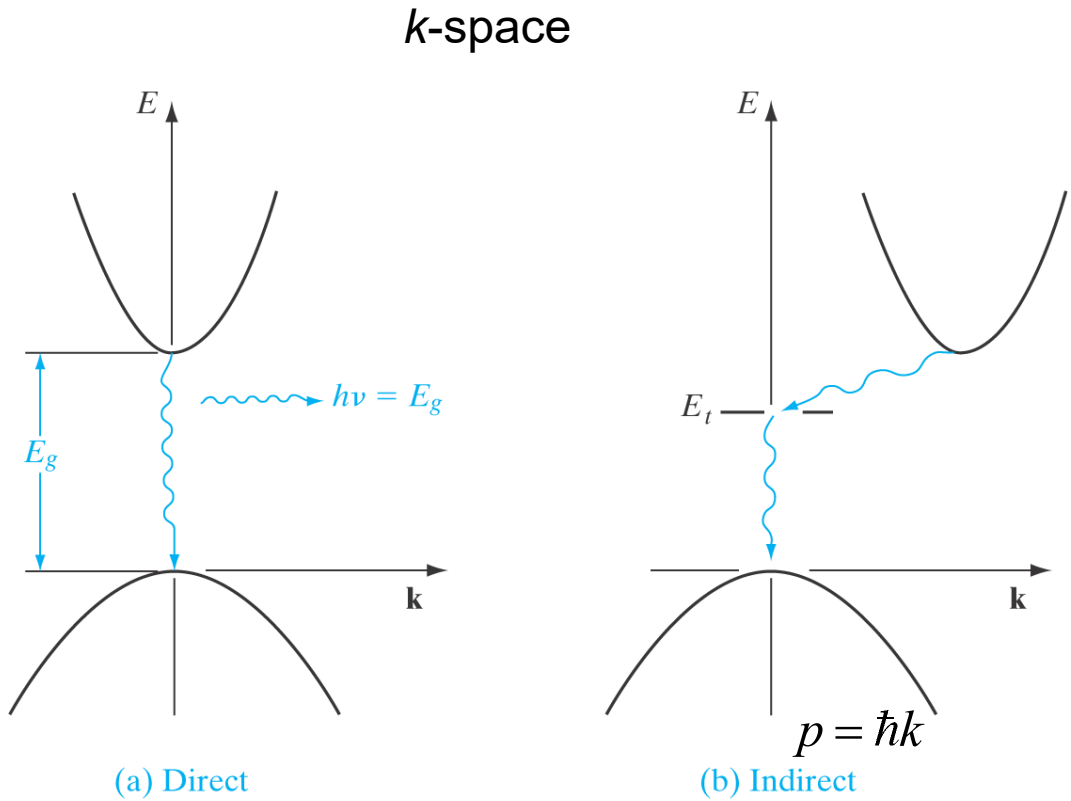
Classical Mechanics: $E = \frac{p^2}{2m}$



What would this look like if the mass were smaller? or negative?

Electron in crystal lattice





k is the wavenumber

$$k = \frac{2\pi}{\lambda} = \frac{p}{\hbar}$$

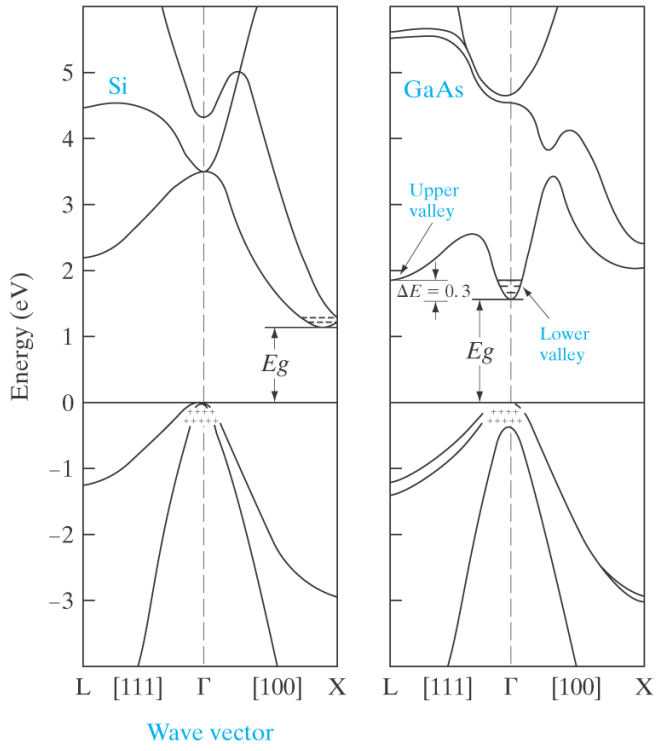
Figure 3.5

Direct and indirect electron transitions in semiconductors: (a) direct transition with accompanying photon emission; (b) indirect transition via a defect level.

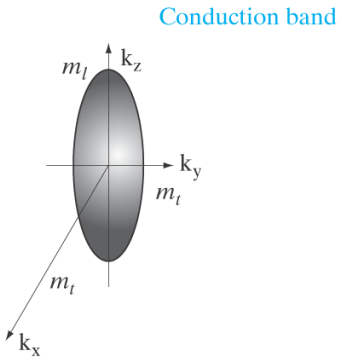
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Si and GaAs in Momentum (k-) Space

For a given k_x, k_y, and k_z the electron has some energy E(eV)



(a)



(b)

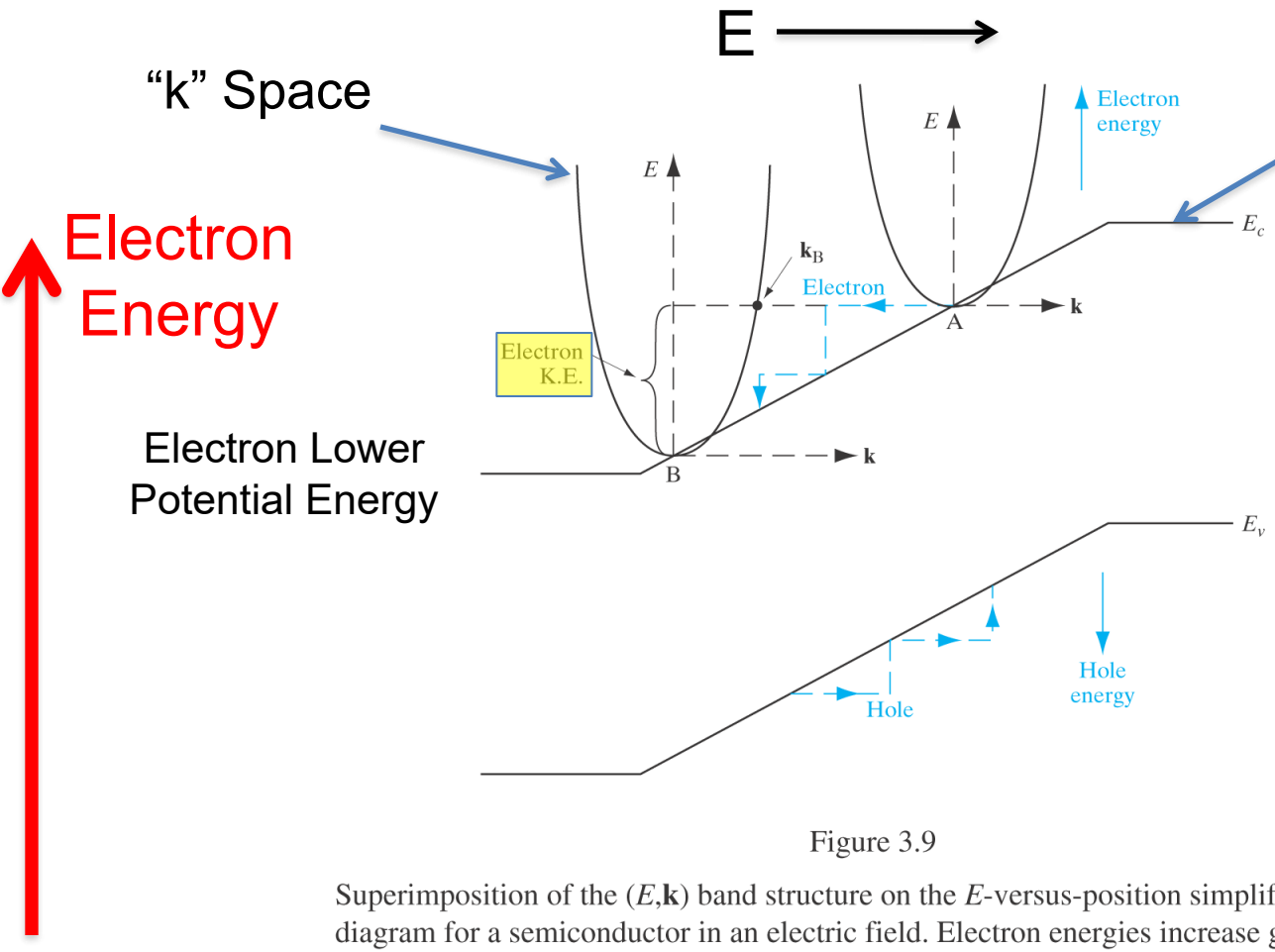
Constant energy surfaces are typically ellipsoids in k-space

Figure 3.10

Realistic band structures in semiconductors: (a) Conduction and valence bands in Si and GaAs along [111] and [100]; (b) ellipsoidal constant energy surface for Si, near the 6 conduction band minima along the *X* directions. (From Chelikowsky and Cohen, Phys. Rev. B14, 556, 1976).

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Inferring Electron Potential & Kinetic Energy From Band Diagrams



- The location of the band edge is an indication of the potential energy
- The location above the band edge is an indication of kinetic energy

Figure 3.9

Superimposition of the (E,**k**) band structure on the E-versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wave vectors point in opposite directions and these charge carriers move opposite to each other, as shown.

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Electrons and Holes

Electrons and Holes in "Perfect" Semiconductors

- When an electron is freed from the valence band where it was involved in atomic bonding, an empty state is left behind
- In a perfect, undoped semiconductor, electron-hole pairs can be created through thermal excitation or by interaction with a photon
- Once created, the electrons and holes are free to move (drift and diffusion)

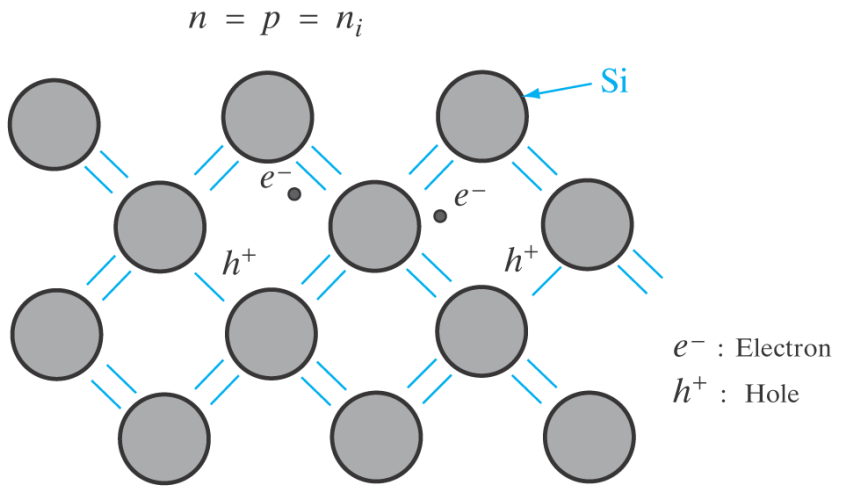


Figure 3.11

Electron–hole pairs in the covalent bonding model of the Si crystal.

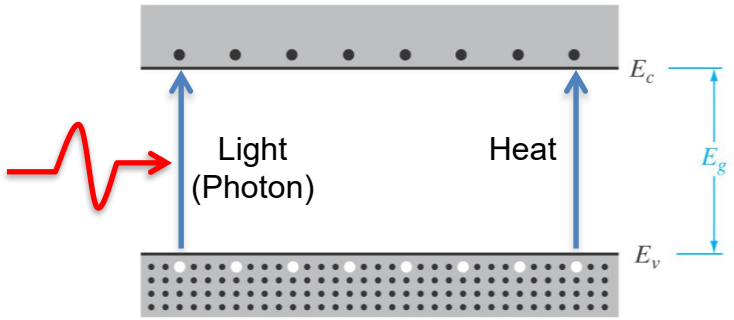
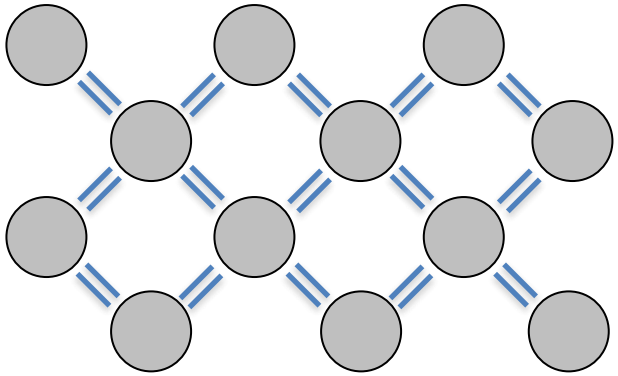


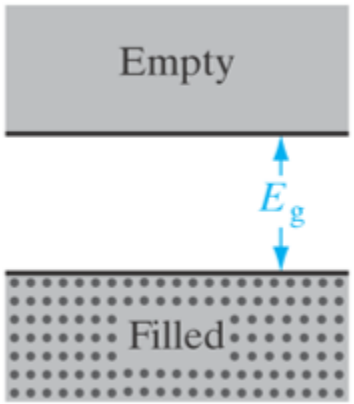
Figure 3.7

Electron–hole pairs in a semiconductor.

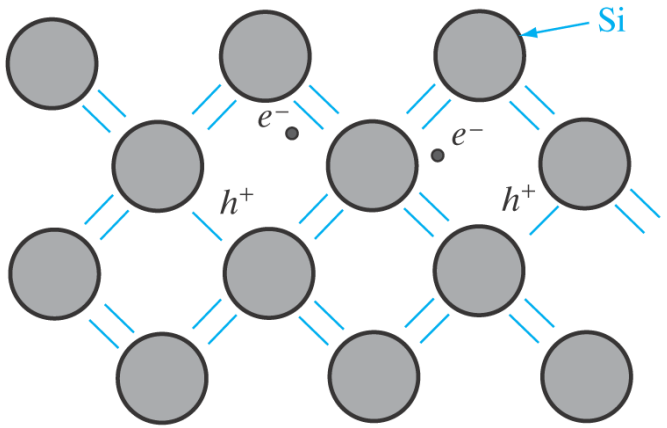
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Electrons and Holes in "Perfect" Semiconductors





$n = p = n_i$



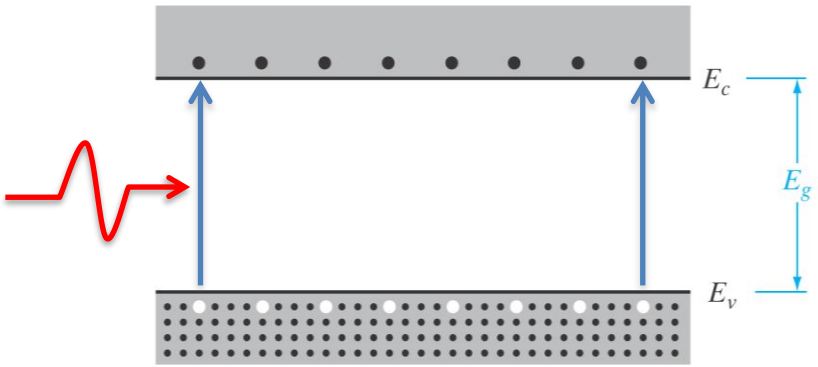


Figure 3.7

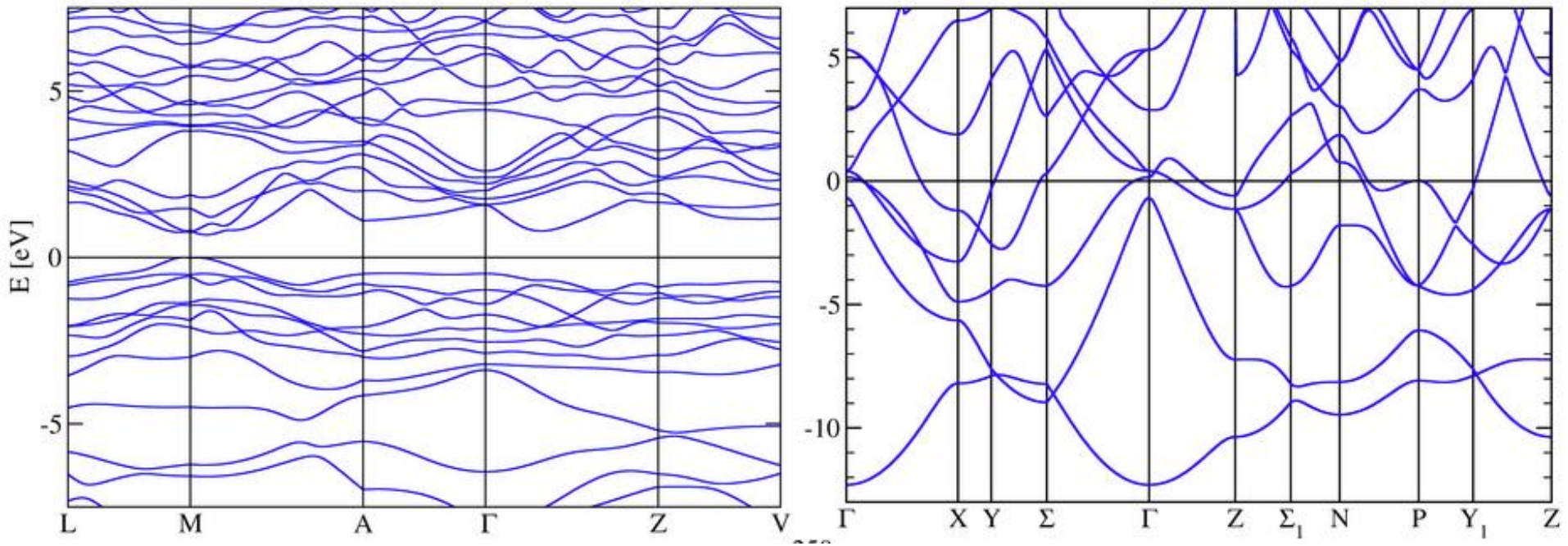
Electron–hole pairs in a semiconductor.

Metals, Semiconductors, and Insulators

Metal and Semiconductor in k-Space

Semiconductor

Metal



- ***Semiconductor/Insulator:*** Maximum valence band energy is lower than minimum conduction band energy
- ***Metal:*** Maximum valence band energy is higher than minimum conduction band energy (band overlap), or partially filled bands

http://www.nature.com/srep/2013/130315/srep01466/full/srep01466.html

Energy Band Picture

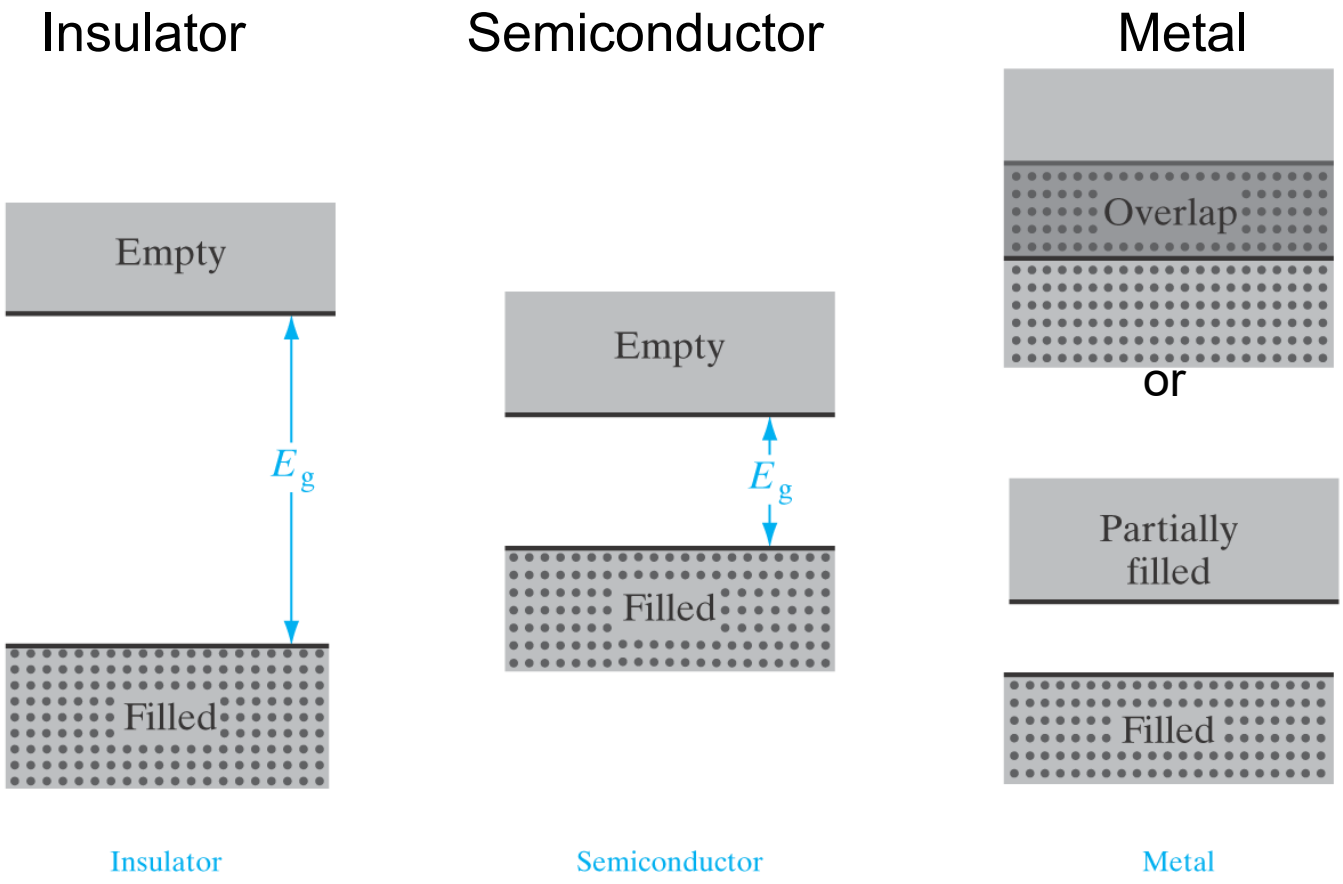


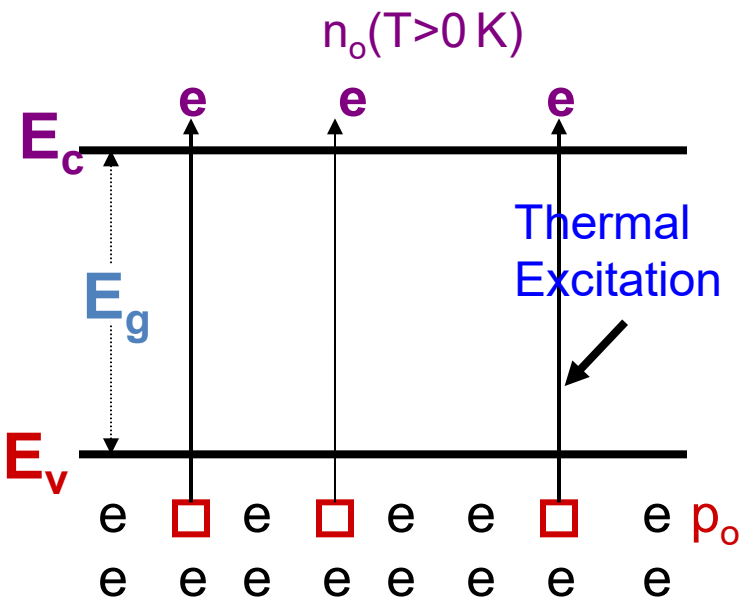
Figure 3.4

Typical band structures at 0 K.

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Definitions

Conduction Band Nx4 States



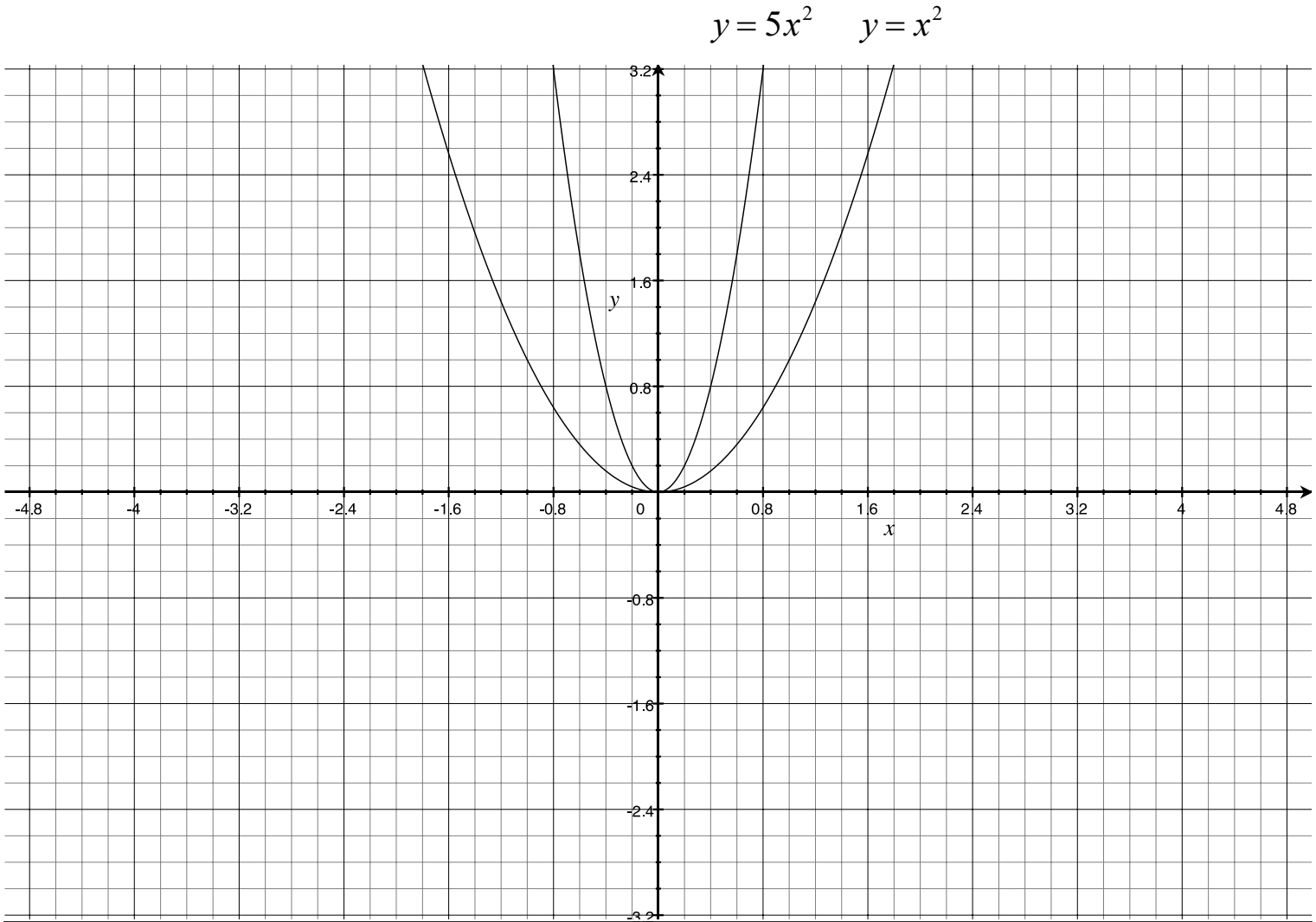
Valence Band: Nx4 electrons occupy Nx4 states at 0 K

- ***Insulator:*** *no electrons are present in the conduction band at T=0 K and the energy gap is large (Eg > ~10 eV). Virtually no electrons can be thermally excited from the valence band to the conduction band at T=300 K.*
- ***Semiconductor:*** *no electrons are present in conduction band at T=0 K and the energy gap is smaller (~0.1< Eg < ~5 eV). Some electrons are thermally excited from the valence band to the conduction band at T=300 K.*
- ***Metal****: Some electrons are present in the conduction band at T=0 K and the conduction band is either partially filled or overlaps valence band.*

Effective Mass

Equation for a Parabola

$y = ax^2$ □ Curvature at origin related to "a" coefficient



Electrons in Free Space

- As a consequence of wave-particle duality, the electron has a wave-like nature

$\psi(z,t) = Ae^{[i(kz-\omega t)]}$

where $k = \frac{2\pi}{\lambda}$ (wave vector); and $\omega = 2\pi\upsilon$ (frequency)

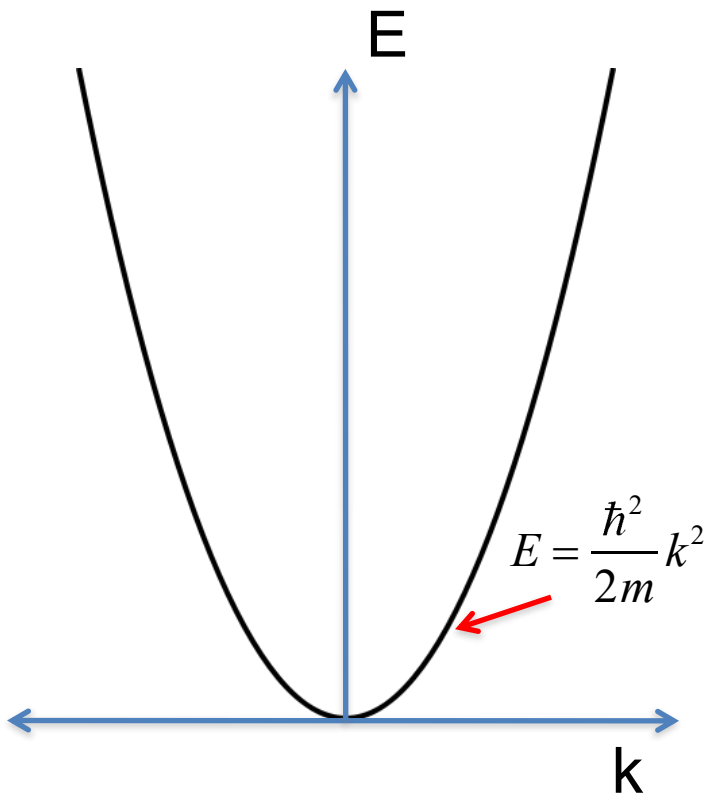
$E = \frac{1}{2}mv^2 = \frac{p^2}{2m}$ (note, here v is velocity)

but $p = \hbar k$

so $E = \frac{\hbar^2}{2m}k^2$

- The E(k) relationship is parabolic for an electron in free space
- The curvature at the origin is related to $\frac{\hbar^2}{2m}$

Classical Mechanics: $E = \frac{p^2}{2m}$



What would this look like if the mass were smaller? or negative?

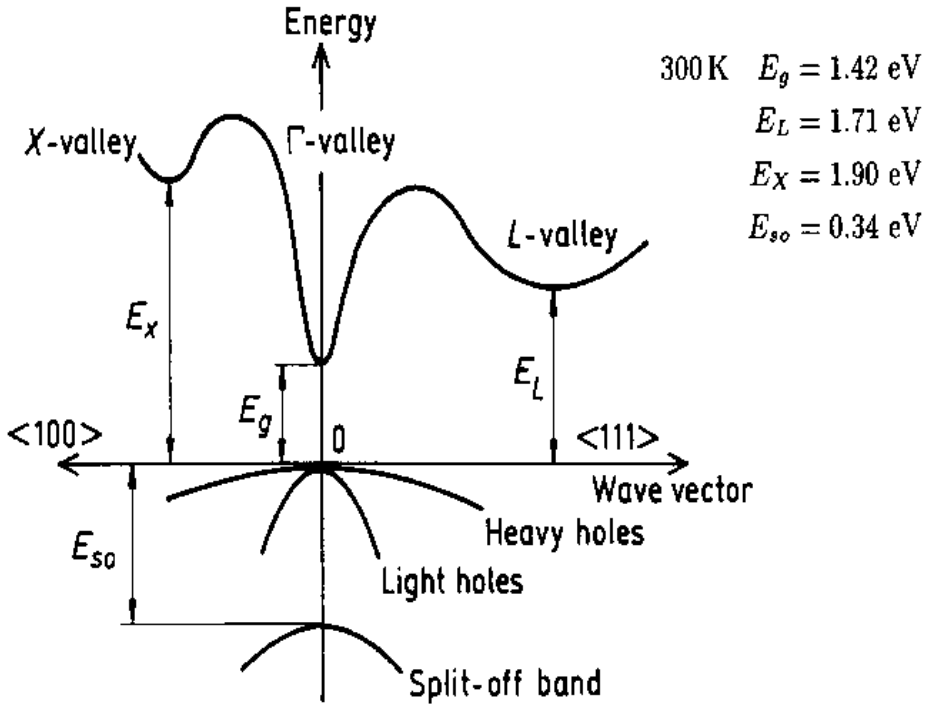
Electron and Hole Effective Mass

- In a semiconductor, the E(k) relationship looks parabolic near the conduction band minima and the valence band maxima
- **The curvature of the parabola differs from the free space curvature**, and depends upon which minima the electron or hole is in
- Near a minima, the E(k) relationship is:

$$E = \frac{\hbar^2}{2m^*}\mathbf{k}^2$$

- This is similar to the E(k) relationship for a free electron, but with a mass m* instead of m_o
- The parameter **m*** is the **effective mass**

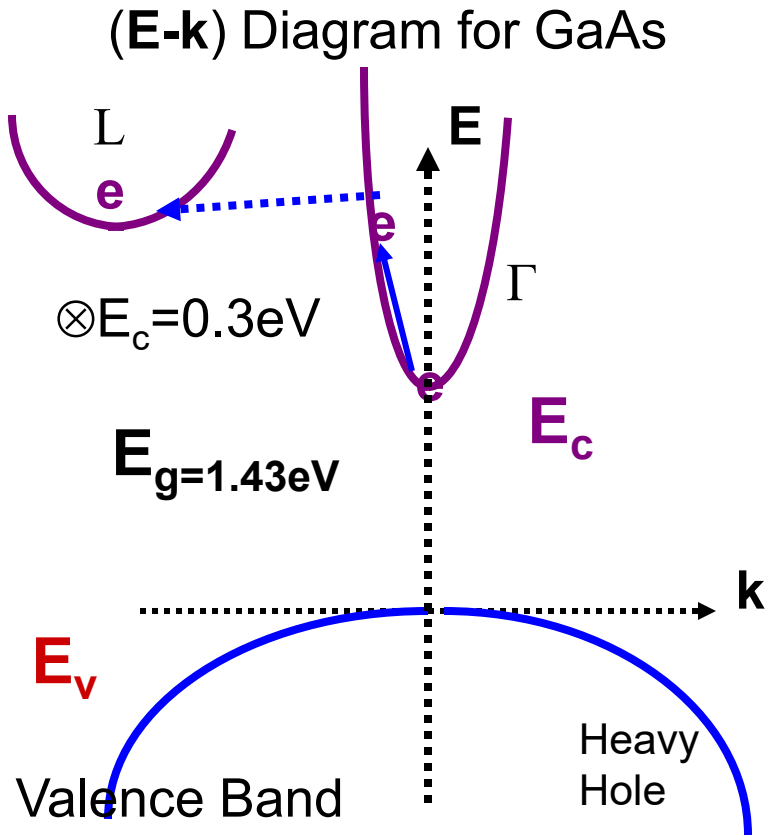
GaAs Band Structure



http://www.ioffe.ru/SVA/NSM/Semicond/GaAs/bandstr.html

Effective Mass in Si, Ge, and GaAs

- In GaAs
  - Lower effective mass electron in Γ Valley
  - Higher effective mass electron in L Valley
- Applied Electrical Field: Effect on Electron
  - Electron gains energy, increases velocity in Γ Valley
  - Electron collides with lattice, emits a phonon, and transfers to the L valley to become a higher effective mass electron



	Ge	Si	GaAs
m_n^*(or m_e^*)	$0.55m_o$	$1.1m_o$	$0.067m_o$
m_p^*(or m_h^*)	$0.37m_o$	$0.56m_o$	$0.48m_o$

A Few Cautions

- The topic of effective mass is more complicated than can reasonably be presented in an introductory course (treatment as tensor in higher level courses)
- **Make sure you know what effective mass to use and how to use it based upon what you are calculating**
- Calculation of the carrier concentration uses the **density of states effective mass**
  - For Silicon, the density of states effective mass is $(m_n^*)^{3/2} = 6(m_l m_t^2)^{1/2}$
  - This comes from the 6 indirect valleys, and the elliptical shape of those valleys
  - Direct-gap materials such as GaAs have a single minima, so just the effective mass is used
- **Conductivity effective mass** is different than the density of states effective mass for Silicon $$m_n^* = \frac{3}{(\frac{1}{m_l} + \frac{1}{m_t} + \frac{1}{m_t})}$$
- **Cyclotron effective mass** is also different
- There is further complication around the direct-indirect crossover in a material where multiple valleys can be populated

What about "Hole" Potential & Kinetic Energy?

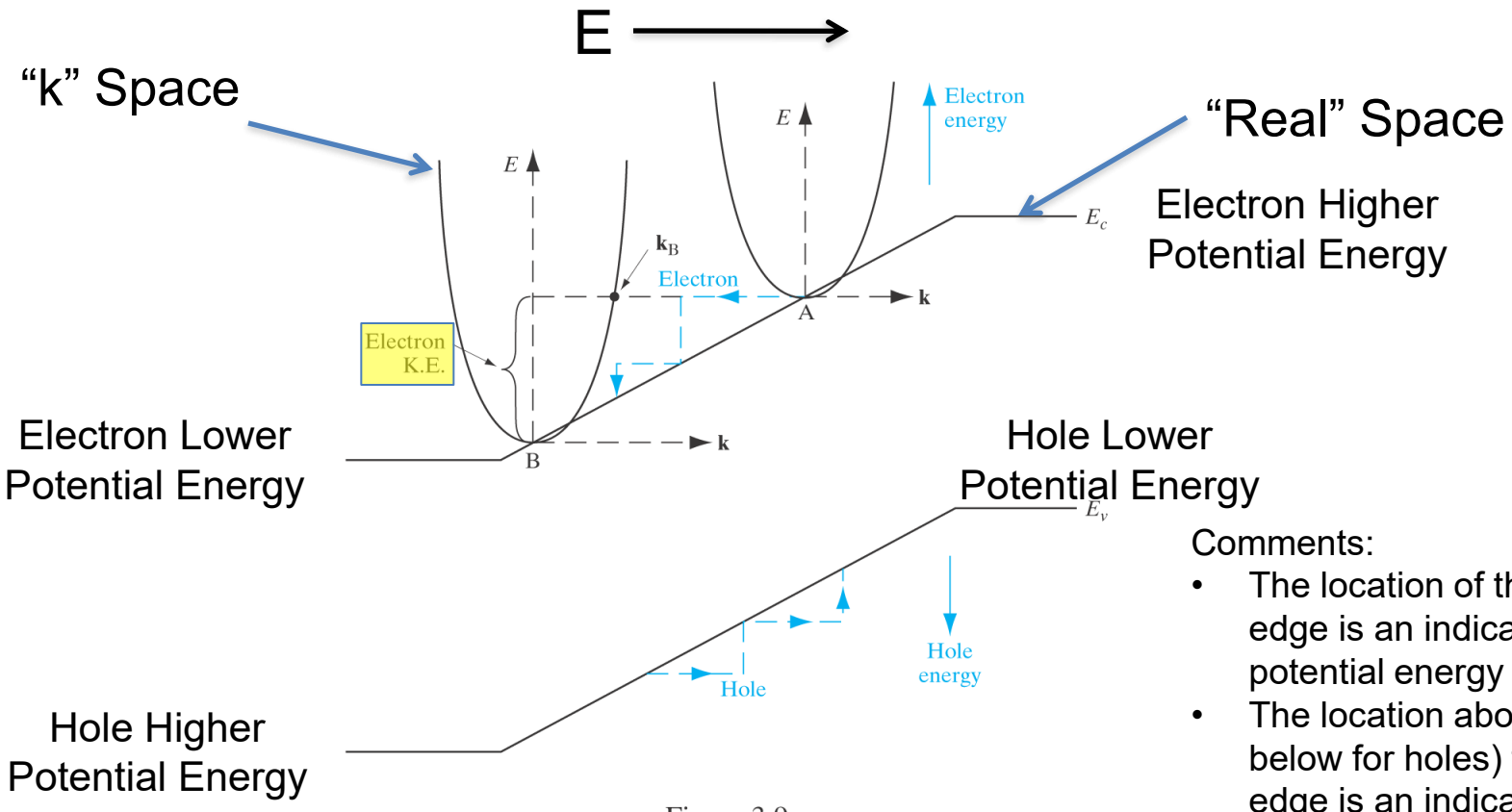


Figure 3.9

Superimposition of the (E,**k**) band structure on the E-versus-position simplified band diagram for a semiconductor in an electric field. Electron energies increase going up, while hole energies increase going down. Similarly, electron and hole wave vectors point in opposite directions and these charge carriers move opposite to each other, as shown.

Comments:

- The location of the band edge is an indication of the potential energy
- The location above (or below for holes) the band edge is an indication of kinetic energy

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